MILITARY SPECIFICATION

FILTER PANEL, INFANT PROTECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a panel of gas-aerosol filter material with protective screening and molded edgeseal.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper (for Carton Sealing).

MILITARY

MIL-P-116 - Preservation Packaging, Methods of.
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-B-51097 - Backing Material, C18.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

FSC 4240

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

C5-53-44 - Filter Panel.
DLB136-42-102-9 - DOP Filter Testing Penetrometer, Q127.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to the specifications and drawings forming a part of this specification.

3.2 Edgeseal configuration. The edgeseal configuration shall meet the requirements of Figure 1.

3.3 Screening. The screening shall be embedded in the edgeseal; however, it is permissible for the screening to show on the outer edge when examined as specified in 4.4.4.4.

3.4 Edgeseal material. The material shall be cured at a minimum temperature of 345°F for a minimum of 3 minutes.

3.5 Backing material. The backing material shall conform to MIL-B-51097 except for water repellency. The average water repellency of the backing material used in the filter panel shall be no less than 16 inches.

3.6 Manufacture. The filter panels shall be manufactured in accordance with Drawing C5-53-44. No mold release compound or lubricant shall be used during the edgeseal molding operation (see 6.3).

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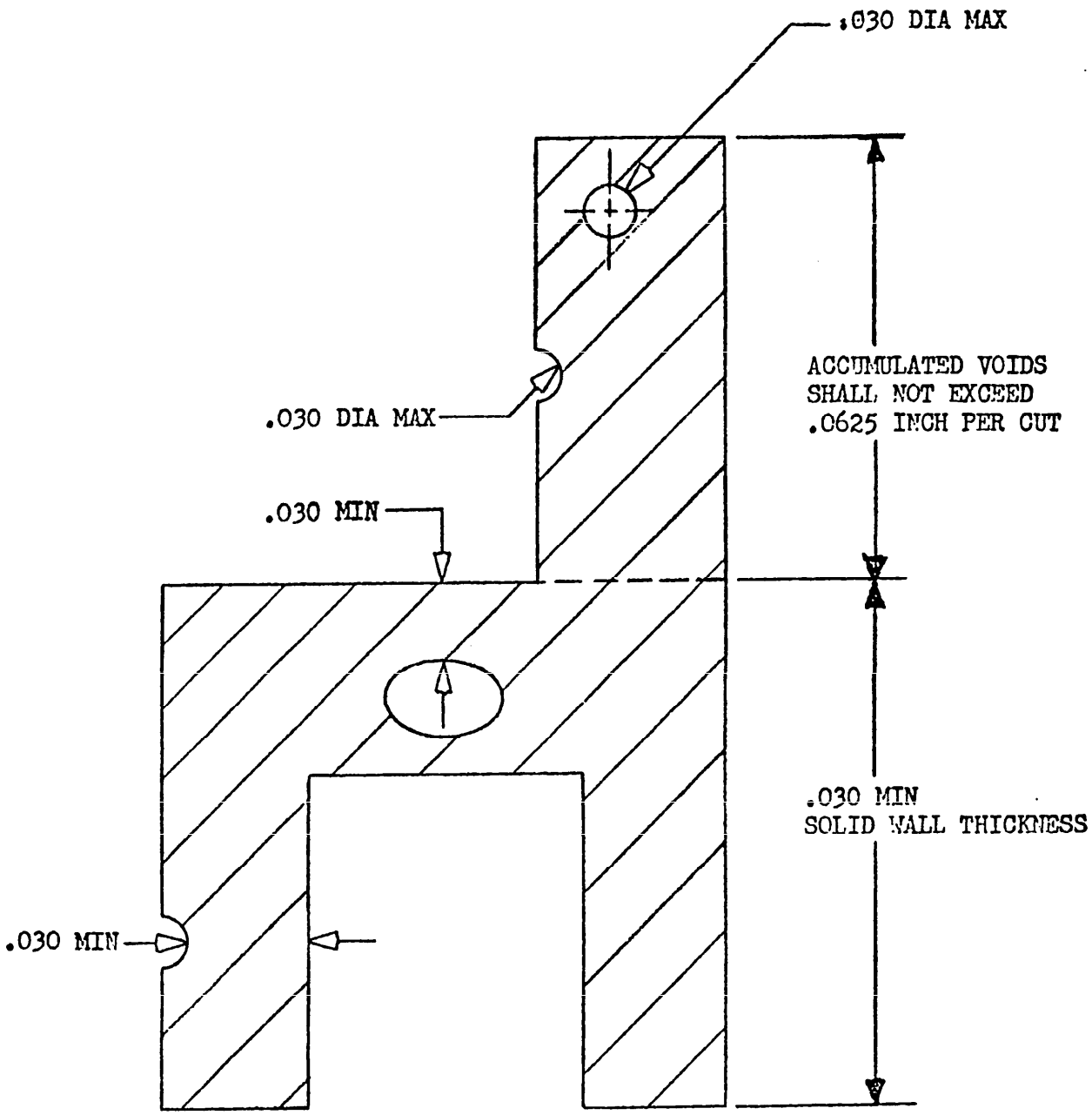


Figure 1

EDGESEAL CONFIGURATION

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3.7 Edgeseal adhesion. The force required to pull the edgeseal material loose from the screening and the filter material shall be not less than 17 pounds when tested as specified in 4.4.4.1.

3.8 DOP smoke penetration. DOP smoke penetration of each filter panel shall not exceed 0.05 percent when tested as specified in 4.4.4.2.

3.9 Moisture content. The average moisture content of the panels after packaging shall not exceed 2.0 percent and no individual panel shall exceed 2.5 percent moisture when tested as specified in 4.4.4.3.

3.10 Water absorption. The water absorption of each filter panel shall not exceed 17 grams, measured in terms of an increase in weight when tested as specified in 4.4.4.5.

3.11 Bondability. The filter panel edgeseal shall be capable of being bonded to flexible polyvinyl chloride film and the peel strength of such bonds shall be not less than 17 pounds when tested as specified in 4.4.4.6.

3.12 Screen heatseal. The heatsealing of the screen to the filter material shall allow the center grid to support the weight of the panel when tested in accordance with 4.4.4.7.

3.13 Preproduction. Prior to the start of regular production a preproduction lot of panels shall be produced in accordance with this specification for examination and test (see 4.3).

3.14 Workmanship. The filter panels shall be neatly trimmed of flash, free from foreign matter, contamination, cuts, or damage and shall show no evidence of having been overheated, wetted, or mishandled. The edgeseal shall be free of surface in excess of those shown in Figure 1. No filter material shall be visible on the surface of the edgeseal. Excess edgeseal material on the filtering surface of the panel shall not exceed a total of 2 square inches in area. The screening shall not pucker, bulge, or be folded over.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisified.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 10 panels from each edge sealing mold cavity or 120 panels, whichever is greater, shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive test. Each filter panel of the preproduction sample shall be inspected and tested for all of the characteristics included in the classification of defects (see 4.4.3.4).

4.3.2.2 For destructive tests. Twenty filter panels for each test, taken at random from the preproduction sample, shall be tested for moisture content, water absorption, bondability, screen heatseal, and edgeseal adhesion, and for edgeseal configuration and screening by sectionalizing the panels for examination.

4.3.3 Acceptance/rejection criteria. The sample filter panels shall meet the examination and tests specified in 4.3.2.1 and 4.3.2.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filter panels produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of filter

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material or one lot of edgeseal material shall be represented in any one lot of finished filter panels.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Filter panels shall be taken in accordance with MIL-STD-105, level S-3. However, the sample size in no case shall be less than 20 filter panels for each test.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample filter panels and the preparation for delivery from the regular production lots shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests.

4.4.3.2.1 For moisture content, bondability, screen heatseal, and edge-seal adhesion. Sample filter panels from the regular production lot shall be tested in accordance with 4.4.4.3, 4.4.4.6, 4.4.4.7 and 4.4.4.1 for compliance with 3.9, 3.11, 3.12, and 3.7 using AQL of 0.65 percent defective.

4.4.3.2.2 For water absorption, edgeseal configuration and screening. Sample filter panels from regular production lot shall be tested in accordance with 4.4.4.5 and 4.4.4.4 for compliance with 3.10, 3.2 and 3.3 using AQL of 2.5 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

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4.4.3.4 Classification of defects.

(a) Filter panel (Dwg C5-53-44).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Plan view profile incorrect	
103	Screen puckered, bulged, or folded over	
104	Filter material dirty or abraded	
105	Marking incorrect	
106	Contamination (grease or oil)	
107	Workmanship (see 3.14)	
Minor:	AQL 2.5 percent defective	
201	Overall width incorrect	
202	Overall length incorrect	
203	Contamination (other than 106)	
204	Marking missing or illegible	

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	
1	Drum leakage (see 5.2)
Major:	AQL 1.0 percent defective
101	Container closure incorrect
102	Container gasket missing or misaligned
Minor:	AQL 2.5 percent defective
201	Packaging or packing component missing, incorrect, damaged, or incorrectly assembled
202	Marking missing, incorrect, or illegible
203	Contamination

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4.4.4 Tests.

4.4.4.1 Edgeseal adhesion. A 1 inch wide specimen shall be cut from each side of the sample filter panel. The specimens shall be of sufficient length to facilitate pull test. The filter material and screening shall be fastened to one jaw of a tensile tester (Scott or equal) and with a suitable fixture fastened to the other jaw to pull against lower edge of edgeseal (junction of edgeseal and panel). Test shall be conducted at a rate of travel of 20 inches per minute.

4.4.4.2 DOP smoke penetration. DOP smoke penetration of the filter panels shall be determined with the Q127 DOP Filter Testing Penetrometer (DLB136-42-102-9), at a flow rate of 32 liters per minute.

4.4.4.3 Moisture content. Moisture content of the filter panels shall be determined as follows:

Weigh each sample panel to the nearest 0.1 gram (wet weight) immediately after unpackaging. Place the panels in a forced air drying oven at 160° ± 5° F and a maximum of 5 percent relative humidity until constant dry weight, within 0.1 gram, is reached. Record the final weighing (dry weight) to the nearest 0.1 gram. Calculate the percent moisture as follows:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{\text{wet weight}}$$

4.4.4.4 Edgeseal configuration and screening. The sample filter panels shall be sectionalized, using a steel rule die or a similar cutting tool, in such a manner that each cut section shall be 1 ± 1/16 inch wide with a 1/4 inch minimum depth. The cut edges shall be perpendicular to the outer wall of the edgeseal. The cut edges shall be clean and straight. Two sections shall be taken at random from each side of the sample filter panels. All sections shall be examined for the requirements of 3.2 and 3.3. Delamination of the filter material under the edgeseal shall not be cause for rejection.

4.4.4.5 Water absorption. Water absorption of the filter panels shall be determined as follows:

(a) Weigh each of the sample filter panels to the nearest 0.1 gram and record as the dry weight.

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(b) Stand the panel upright on one of the long edges in a suitable dish, and add room temperature distilled water to the dish until the water is 1 ± 1/16 inch deep.

(c) Allow the panels to stand in the water for 30 ± 1 minute.

(d) At the end of 30 minutes, remove the panels from the dish of water and remove all visible surface moisture with a cloth or towel. The panels should appear to be dry.

(e) Within 10 minutes after removal from the water, weigh each of the sample panels again to the nearest 0.1 gram and record as the wet weight.

(f) Subtract the dry weight from the wet weight for each of the sample elements to obtain the water absorption.

4.4.4.6 Bondability. The test specimen shall be prepared by heat sealing a sheet of flexible polyvinyl chloride (PVC) film to each of the four sides of the sample filter panel with the Government approved dielectric heat sealing equipment using the following procedures and parameters:

(a) Place panel under press so that the 1/4 inch wide panel lip is under the 1/4 inch by 8 inches press electrode. Position so that the panel side is approximately centered with the electrode length and the electrode seats on the panel lip without striking the 5/16 inch vertical panel wall.

(b) Lay 6 inches by 10 inches PVC sheet centered in press on top of panel. The vinyl film will be from a controlled lot accepted as suitable for bonding.

(c) Sealing conditions as itemized below:

(1) Generator frequency = 27.12 ± 0.16 megahertz
(2) Temperature of electrode sealing surface (0.25 inch by 8 inches) = 260° to 320° F.
(3) Press cylinder = 3.5 inches diameter
(4) Sealing air pressure = up to 5 pounds per square inch (psi)
(5) Squeeze air pressure = 20 ± 2 psi
(6) Preseal time = 2 seconds ± 5 percent
(7) Seal time = 3 seconds after preseal time ± 5 percent

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(8) Squeeze time = 2.75 seconds after preseal time ± 5 percent

(9) Down-Stop = 0.093 ± 0.002 inch gap between electrode and lower platen

(10) Cool time = 2 seconds ± 5 percent

(11) Radio Frequency (RF) voltage between upper and lower platen = 1.00 to 2.50 kilovolts peak to ground.

(12) Lead length = 10 inch with power control set to obtain final sealed thickness 0.100 ± 0.010 inch

(13) Arc Suppressor sensitivity to allow full seal cycle.

(d) After minimum of 15 minutes from seal time, cut 1 inch wide strips from center of each sealed side. Conduct a peel test by placing the 1 inch wide strip ends to be gripped in the jaws of the testing machine. The bonding peel strength shall be determined in accordance with method B of method 1013 of Fed. Test Method Std. No. 406 using a pendulum (Scott) type of machine except that no autographic stress strain curve is needed; the rate of travel shall be 20 inches per minute.

4.4.4.7 Screen heatseal. Cut the screen on the panel as shown in Figure 2. Hang the panel using hooks from points A, B, C, and D for 30 seconds. Repeat the test on the opposite side. Peel the cut section of the screen from the filter material and the entire center grid must show evidence of attachment to the filter material. The attachment shall be indicated by nap from the filter material being bonded to the screen.

4.4.5 Acceptance/rejection criteria. The sample filter panels shall meet all the examination and tests specified in 4.4.3.1, 4.4.3.2 and 4.4.3.3 to be acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Thirty filter panels (Dwg C5-53-44) shall be packaged in fiberboard box conforming to style RSC, type CF, variety SW, class-domestic, grade 200 of PPP-B-636. The box shall be small enough to fit in a horizontal position inside the 55-gallon container specified in 5.2. Filler pads of the same material as the box shall be used as required to ensure a tight package. Three inch wide tape conforming to PPP-T-76 shall be used to close and seal the box by running the full length of the seams and extending over the ends not less than 2-1/2 inches.

5.2 Packing. Three packages, from 5.1, shall be packed in a 55-gallon drum conforming to type IV of PPP-D-729. Desiccant, type II, 115 units, conforming to MIL-D-3464 shall be equally distributed inside the drum and secured in place to prevent movement and rupture of the bags. Pads or cells of fiberboard conforming to type CF, variety SW, class-

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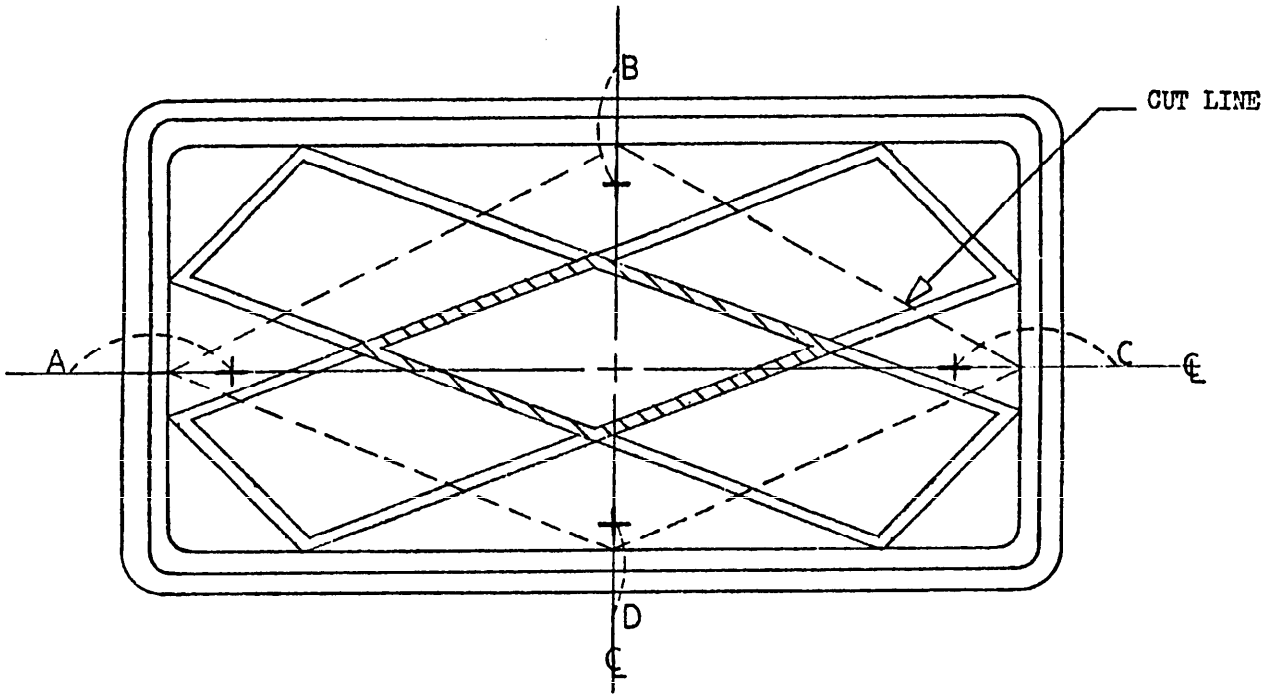


Figure 2

CUTTING FOR SCREEN HEATSEAL TEST.

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domestic, grade 200 of PPP-F-320 shall be formed and used to assure a tight pack. A loosely applied loop of twine, with a minimum breaking strength of 70 pounds shall be secured around each box to facilitate removal. Ends of the twine shall be securely tied in forming the loop. The drum shall be closed to ensure a tight seal when tested in accordance with the leak test for large containers in MIL-P-116 except that all drums shall be tested. The water temperature shall be 50° ± 5° F greater than the temperature of the drum at the time of testing.

5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers filter panels intended to be used in the M2 Infant Protector.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Precautions. Care should be exercised during cutting and handling of the filter material and filter panels to avoid punctures, cuts, surface abrasions, and loss of charcoal from the edges, and to minimize moisture pickup during the entire manufacturing process.

6.4 Referee standards. During preproduction and subject to later revision during production, the Government will establish referee standards to define the type and category (major and minor) of each defect. This classification will be based upon the defect magnitude and potential deleterious effect on intended use.

6.5 CK life. For maximum CK life of filter panels, the panels should be left in original sealed shipping container and should not be heat sealed into the M2 Infant Protector until time of issue.

6.6 Moisture. For minimum moisture pick-up of filter panels, all packaging and packing materials should be dried prior to use.

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6.7 Component specifications.

Edgeseal Material, Chemical-Biological Filter	MIL-E-51420
Filter Material, Gas Aerosol, C18R1	MIL-F-51095
Backing Material, C18	MIL-B-51097
Screening, Chemical-Biological Filter	MIL-S-51421
Core Material, C18	MIL-C-51096
Charcoal, Activated, Impregnated, ASC	MIL-C-13724

6.8 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-F-1060B, dated 14 March 1972.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A668

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the [illegible] of the specification in procurement of products for ultimate use by the Department of Defense. This sheet [illegible] provided for obtaining information on the use of this specification which will insure that suitable products [illegible] be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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